

Continue â€” Project Status Snapshot

Revision: 17 **Last modified:** 2026-06-10T11:55:33Z **Session:** 2026-06-10 (Session 11 â€” BobaLink browser extension: Phases 2 & 3 + shell + background SW + Phase 4 api leaf landed; 6-stream parallel session in flight) **Last commit:** 15a9a61 (Phase 3 capstone â€” background service worker message router) **Branch:** main **Working tree:** CLEAN (after the doc commit that carries this update)

Send continue to pick up exactly where we left off. This file is the single source of truth for session handoff (Â§12.10 / Â§11.4.131).

CURRENT STATE â€” Session 10 (BobaLink browser extension)

New major feature in flight: the BobaLink browser extension. Plan + discovery + Phase 1 foundation + backend BE-1/BE-2 + a token-auth security fix landed. The Python unit suite stays FULLY GREEN at 4149 passed.

What is BobaLink

A WXT + TypeScript Manifest-V3 browser extension that detects magnet links and .torrent URLs on any page and forwards them to the running Boba merge service on port **7187**. Backend contract: POST `http://<host>:7187/api/v1/download`. The full plan, discovery analysis, and traceability live under `docs/browser_extension/` â€” master plan IMPLEMENTATION_PLAN.md (9 phases), analysis artifacts `_analysis/01â€”06`, planning artifacts `_plan/Aâ€”F`, and the 245-item traceability matrix `_plan/C-traceability-matrix.md` (240 v1 / 5 v2, proving nothing is skipped per Â§11.4.118).

What landed (Session 10 â€” BobaLink)

- b2356ae â€” BobaLink master implementation plan + discovery/planning artifacts (`docs/browser_extension/`); also bumped HelixQA submodule bcac236 â†’ 4d2dcb2.
- 33a9815 â€” extension **Phase 1 foundation** scaffolded (`extension/` â€” WXT config, TS, shared libs/types under `extension/src/`); **71 anti-bluff Vitest unit tests** (vitest 71 passed, eslint clean â€” crypto imports the REAL module, not an inline copy).
- d46bffb â€” backend **BE-1** (CORS for extension origins) + **BE-2** (raw .torrent upload endpoint) added to the Python :7187 API.
- 284d1c4 â€” bumped HelixQA submodule 4d2dcb2 â†’ bca3b36; new Challenge bank submodules `helixqa/banks/boba-bobalink.yaml` (6 cases).
- 192b945 â€” **security fix**: env-gated shared-secret BOBA_API_TOKEN enforced on the three :7187 download-write endpoints (when the env var is set, requests must carry a matching X-Boba-Token / Bearer token; behaviour unchanged when unset). Verified by **15 token-auth tests** (`tests/unit/api_layer/test_download_token_auth.py`).

What landed since Session 10 (BobaLink â€” Phases 2 & 3 + Phase 4 leaf)

- 7225470 â€” Phase 2 wave-1: parsers (bencode.ts/magnet.ts) + scanner base/site-db.
- fa03323 â€” Phase 2 wave-2: .torrent-file SHA-1 infohash + link/text scanners + perf/stress.
- 946c61e â€” Phase 2 complete: scanner orchestrator (cross-scanner dedup).
- 2e59572 â€” lint test files clean; lint script extended to tests/.
- e8fde43 â€” Phase 3 shell + Phase 4 api leaf: api / {boba-client, queue} + content/popup/options.
- 15a9a61 â€” Phase 3 capstone: background service worker (message router). **HEAD.**

Suite status

- Full Python unit suite: **4149 passed** â€” `pytest tests/unit/ --import-mode=importlib` (Session 10 baseline; unchanged this session). Evidence: `qa-results/tokenauth_fullsuite_*.log`.
- Extension Vitest corpus: **379 passed across 33 spec files** â€” same-session `npx vitest run (Tests 379 passed (379)); tsc --noEmit clean; npm run lint 0/0.` (+92 over the 287/22 baseline.) Build: `npx wxt build â†’ loadable .output/chrome-mv3/` (Â§11.4.38 â€” 8/8 manifest assets verified present).

Current phase + next steps (Session 11 â€” two parallel waves landed; reviewed checkpoint)

- **Phases 1, 2, 3 + extension shell + background SW: COMPLETE.**
- **WXT build wiring: COMPLETE** â€” endpoints at `src/entrypoints/{background.ts, content.ts, popup/index.html, options/index.html}` (thin wrappers); `npx wxt build â†’ loadable .output/chrome-mv3/`; matches derived from `SITE_SELECTORS` (24 hosts, no `<all_urls>`); least-privilege manifest; Â§11.4.38 verified.
- **Phase 4 (backend integration): IN PROGRESS** â€” api leaf @e8fde43; **Phase-7 decrypt-before-send WIRED** â€” `BobaClient.create()` decrypts the encrypted `BobaApiToken` bundle; background reads the session passphrase from `chrome.storage.session`, sends decrypted plaintext, default-open when locked (REDâ†’GREEN; token/passphrase never logged). **PENDING:** live-7187 integration (`require_backend(7187)`) + detectâ†’sendâ†’torrent-in-qBittorrent E2E on the real backend.
- **Phase 5 (tab groups): IN PROGRESS** â€” standalone `src/tabgroups/index.ts` (dedupe-across-group + batch dispatch) + 13 tests landed; **PENDING:** wire into background `MENU_SEND_GROUP` + add `tabGroups/tabs` perms (least-privilege review first).
- **Phase 6 (i18n): IN PROGRESS** â€” `locale.test.ts` guards en-catalog completeness. **Phase 7 (security): IN PROGRESS** â€” decrypt path + `tests/security/*` (least-privilege/CSP/no-hardcoded-secret/secret-storage). **Phase 8: IN PROGRESS** â€” 379/33 green; real-module Challenge challenges/extension/detect_and_forward_challenge.sh (mutation-verified); HelixQA BOBA-LINK-007; e2e is an honest operator-gated SKIP (sandbox canâ€™t load extensions). **Phase 9: PENDING** â€” `ci-ext.sh` gate + per-store packaging + Â§11.4.65 doc siblings.
- **Next actions:** (1) **independent-subagent code-review re-pass** for the correctness/security/build lenses once the platform subagent-dispatch throttle clears (done in-context this session; re-run before any release tag per Â§11.4.40); (2) Phase 5 integration into

background + perms; (3) Phase 4 live-7187 integration + E2E; (4) Phase 9 packaging; (5) regenerate Status_Summary.md + Â§11.4.65 HTML/PDF siblings for the browser_extension Status docs. Status: docs/browser_extension/Status.md (Rev 2, accurate). QA evidence: docs/qa/bobalink-2026-06-10-session11/.

CURRENT STATE â€” Session 10 (2026-06-10, subagent-driven loop)

Suite stays **FULLY GREEN**. This session was **hardening + verification, subagent-driven** (Â§11.4.70), every change code-reviewed before commit (Â§11.4.125/Â§11.4.134).

What landed (Session 10)

- `test_credential_env_wiring.py` **isolation fix** â€” resolved the Session-9 â€œknown low-priority follow-upâ€” below. A `merge_service` fake-namespace stub was installed at module-collection time with no teardown, so `confTest`'s `_isolate_download_proxy_modules` captured-and-restored the *fake* stub into siblings in some run orders. Fixed by moving stub install into a `search_mod` fixture that snapshots/restores the 3 injected keys (`merge_service`, `.search`, `.retry`) in finally. **Evidence:** RED reproduction (the 5 credential tests + a temporary leak-probe) went 1 failed â†’ all green; the fileâ€™s own **5 tests pass** standalone; full unit suite **4121 passed** — seeds default/42/31337 (Session-10 run). Code-review **GO** (proved negation; no weakened assertions; ruff clean).
- docs/qa/BOB-008/operator_runbook.md â€” copy-pasteable operator unblock procedure for BOB-008 (cookie path preferred; CAPTCHA path documented with its `cap_sid/cap_code_field` friction), every endpoint claim cited to `auth.py:line` (Â§11.4.83/Â§11.4.99).
- **Latent-leak discovery sweep** (Â§11.4.118) â€” audited all of `tests/` for the module-level fake-stub-no-teardown pattern. Finding: the pattern is a **structural no-op wherever the test dir has an `__init__.py`** (real package pre-loads â†’ `setDefault` never installs the fake), so the ~26 `tests/unit/merge_service/` matches and `test_tracker_stats.py` are **inert** (no RED reproduces; strict TDD â†’ no fix). The single genuinely-uncovered root is `pirateiro` (see follow-up).
- **Constitution inheritance verified** (Â§11.4.32/Â§11.4.35) â€” PASS: pointer present in both layers, propagated anchors present in canonical source, no conflict markers, `pre_build_verification.sh` 18/18 green.
- **pirateiro test-isolation fix** â€” **BOB-063 (operator-approved, implemented)** â€” `tests/unit/test_plugin_pirateiro.py` injected `sys.modules['pirateiro']` at module scope with no teardown; `pirateiro` was the one root uncovered by `confTest`'s isolation, so it leaked into later tests. The naive â€œadd to `_POLLUTING_ROOTS`â€” does NOT work (the leak is already inside each per-test snapshot); the real fix caches+re-registers+purges the stub per unit test. **Evidence** (Â§11.4.115): RED 1 failed, 44 passed â†’ GREEN 45 passed; negation proof (disable the purge â†’ re-fails); full suite **4122 passed** — seeds default/42. Standing regression guard `tests/unit/test_pirateiro_isolation_guard.py` (Â§11.4.135). QA at docs/qa/BOB-063/evidence.md. **Closed BOB-063** (Task â†’ Completed).
- **Constitution submodule advanced** 60e2d66 â†’ f26368b (Â§11.4.26, **clean fast-forward, Â§11.4.113-safe**) â€” pulled Â§11.4.140 v2 (BACKGROUND action), Â§11.4.142 (universal code-review â€” already actively enforced this session; added to

CLAUDE.md propagated clauses), Â§11.4.143 (video-streaming real-user-journey â€” latent/N-A to this torrent-proxy project). Cascade gate **18/18 green** at the new pin.

Open queue (Issues.md)

- **BOB-008** â€” RuTracker CAPTCHA â€” **OPERATOR-BLOCKED**. Unblock procedure documented at docs/qa/BOB-008/operator_runbook.md (preferred: paste bb_session via POST /api/v1/auth/rutacker/cookie-login).
- Everything else closed (DBâ†”i,ŽMD in sync, **63 items**, bin/workable-items validate OK).

Known low-priority follow-up

- ~~pirateiro test isolation defense in depth~~ â€” **RESOLVED this session** (BOB-063, see above).
- ~~test_credential_env_wiring / env_loader stub teardown quirk~~ â€” **RESOLVED this session** (see above).

CURRENT STATE â€” Session 9 (2026-06-10, overnight autonomous loop)

The unit suite is now **FULLY GREEN** and **DETERMINISTIC**.

Metric	Value
Unit tests	4121 passed, 0 failed â€” pytest tests/unit/ --import-mode=importlib (5m24s)
Determinism (Â§11.4.50)	top-level scope 3150 passed identical across --randomly-seed 7/42/100/31337/12345
Commits this session	6e15a8d (crash guards + async/loop hangs), 6230865 (test-pollution + network timeouts) â€” both pushed to all upstreams
Code review	GO (zero findings/warnings) after Â§11.4.134 iterate-until-GO

What landed (Session 9)

- **8 product fixes** (6e15a8d): degenerate-input crash guards for tokyotoshokan/kickass/yts/piratebay; enricher full-suite-hang fix (aiohttp.ClientTimeout Â—6); kickass/bitsearch/torrentgalaxy unbounded-loop caps (MAX_PAGES=50); mutation-scanner .venv scope fix; +7 download_proxy tests. Closed **BOB-060**.
- **Test-suite stabilization** (6230865): eliminated all Â§11.4.50 order-dependent pollution â€” tests/conftest.py _CORRECT_MS_PATH repo-root fix (the dominant bug: corrupted merge_service.__path__ â†” broke 11 tests incl.Â all scheduler_api), extended _POLLUTING_ROOTS, + per-test socket.socket & os.environ snapshot/restore. Network-I/O timeout hardening (search.py/routes.py/helpers.py/eztv.py). Closed **BOB-061**, **BOB-062**.

Open queue (Issues.md)

- **BOB-008** "RuTracker CAPTCHA" **OPERATOR-BLOCKED**: needs you to complete the CAPTCHA at `/api/v1/auth/rutacker/captcha+ /login`, OR paste a fresh `bb_session` cookie via `/auth/rutacker/cookie-login`. Cannot be solved autonomously.
- Everything else closed (DB "i,ŽMD in sync, 62 items, bin/workable-items validate OK).

How to run the suite (both work now)

- Monolithic: `.venv/bin/python -m pytest tests/unit/ -q --import-mode=importlib` 4121 passed.
- Per-scope (faster, parallelizable): `tests/unit/merge_service/` (801) `tests/unit/api_layer/` (170) `tests/unit/*.py` top-level (3150).
- Always pass `--import-mode=importlib` (the `merge_service.deduplicator` lazy import needs it; a bare per-file run errors).

Known low-priority follow-up (NOT blocking " suite is green)

- A residual `test_credential_env_wiring / env_loader sys.modules-stub` ordering quirk surfaces only in narrow isolated file-combos; it does NOT manifest in the full top-level scope (5 seeds green) because the new `env/sys.modules` isolation covers it there. Tighten the `env_loader` stub teardown if it ever resurfaces.

Session 8 Summary

Metric	Value
Test files	47 test modules (222 <code>test_*.py</code> files total across all suites)
Test cases	4,074+ passed
Coverage (unit)	88%+ (gate: 49%)
Bugs fixed	21 (BOB-001 through BOB-025, minus BOB-008)
Submodules aligned	constitution, challenges, containers, helixqa, jactett
Ruff / Mypy	pre-existing warnings only, 0 new

Bug Tracker

21 bugs fixed across Session 8: - 10 B-substring parsing fixes - 3 `import re` missing fixes - 2 comma-size parsing fixes - 2 crash guards - 1 bt4g hang fix - 1 dedup circular import fix - 1 `env_loader` flaky test fix - 1 pirateiro full test suite (44/44)

1 remaining: BOB-008 (RuTracker CAPTCHA) "operator-blocked, needs manual cookie paste.

Test File Inventory

tests/unit/merge_service/ (31 files)

- test_cors_config.py
- test_dead_tracker_bucket.py
- test_deadline_tunable.py
- test_deduplicator.py
- test_deduplicator_edge.py
- test_diag_no_stale_leakage.py
- test_edge_case_challenges.py
- test_enricher.py
- test_enricher_edge.py
- test_hooks.py
- test_html_parsers.py
- test_jackett_autoconfig.py
- test_nnmclub_session_login.py
- test_private_tracker_html_fixtures.py
- test_public_plugin_harness.py
- test_public_plugin_harness_broad.py
- test_public_tracker_capture.py
- test_public_tracker_subprocess_timeout.py
- test_quality_detection.py
- test_scheduler.py
- test_scheduler_coverage.py
- test_search_concurrency.py
- test_search_coverage.py
- test_search_deep_coverage.py
- test_search_error_paths.py
- test_session_encryption.py
- test_theme_endpoint.py
- test_tracker_stats.py
- test_ttl_caches.py
- test_validator.py
- test_validator_coverage.py

tests/unit/api_layer/ (14 files)

- test_auth_coverage_extra.py
- test_concurrent_writers.py
- test_cors_config.py
- test_hooks_coverage.py
- test_hooks_endpoints.py
- test_hooks_remaining.py
- test_nnmclub_auth_endpoints.py
- test_routes_coverage.py
- test_sse_disconnect.py
- test_sse_token_auth.py
- test_theme_state_coverage.py
- test_theme_stream.py
- test_tracker_stats_sse.py

tests/unit/ (55 files, infrastructure + plugin test
test_auth.py, test_auth_coverage.py, test_auth_models.py
test_config.py, test_main.py, test_dashboard.py)

test_content_type_refinement.py
test_credential_env_wiring.py
test_download_proxy_coverage.py, test_download_proxy_deep.py
test_download_merged.py
test_env_loader.py
test_freeleech.py
test_graceful_shutdown.py
test_helpers.py
test_log_filter.py
test_merge_trackers.py
test_novaprinter.py
test_private_tracker_search.py
test_public_tracker_subprocess.py
test_retry_policy.py
test_routes.py, test_routes_coverage.py
test_scheduler_api.py
test_sorting_weights.py
test_sse_disconnect.py
test_streaming.py
test_theme_injector.py, test_theme_wiring.py
test_webui_theme_injector.py
test_ui_module.py, test_ui_sorting.py
test_plugin_*.py

(39 plugin modules: eztv, piratebay, s
limetorrents, torlock, gamestorrents,
anilibra, torrentgalaxy, yts, rutor,
snowfl, torrentdownload, linuxtracker
nmclub, rutracker, megapeer, jackett
one337x, extratorrent, torrentfunk, t
therarbg, academictorrents, ali213, y
glotorrents, pctorrent, rockbox, bitr
torrentscsv, xfsb, yihua, pirateiro,

test_*_guards.py, test_*_deep.py (crash guards + deep coverage variants
test_ci_infra.py, test_ci_workflows.py
test_community_plugins_compile.py
test_courses_scaffold.py, test_course_scripts_lint.py
test_docs_presence.py
test_frontend_spec_coverage.py
test_install_plugin_json_config.py
test_jackett_integration.py, test_jackett_plugin_pool.py
test_nmclub_config_selfheal.py, test_nmclub_plugin_login.py
test_no_runtime_service_skips.py
test_openapi_frozen.py
test_page_title.py, test_footer_restored.py
test_palette_catalog.py, test_palette_catalog_python_mirror.py
test_quality_compose.py, test_quality_detection.py
test_readme_landing_page.py
test_runtime_requirements_includes_new_deps.py
test_scan_script_non_interactive.py
test_scanner_configs.py
test_service_fixtures.py
test_shadow_tokens.py
test_socks_udp.py
test_start_sh_copy_plugins_framework.py
test_toolchain_config.py

```

test_tracker_validator.py
test_website_config.py
test_branding_assets.py
test_architecture_diagrams.py
test_build_releases_non_interactive.py
tests/benchmark/           (3)
tests/chaos/                (2)
tests/concurrency/         (3)
tests/contract/            (5)
tests/docs/                (3)
tests/e2e/                  (7)
tests/integration/         (19)
tests/memory/              (1)
tests/observability/       (2)
tests/performance/         (1)
tests/property/            (2)
tests/security/            (10)
tests/stress/              (2)

```

Commits (latest 10+)

```

c8254ee fix: 21 test failures, pirateiro 44/44, coverage gate 88%, dedup c
ccd0dc9 fix: env_loader flaky test + coverage baseline 88.14% + minor test
3d1819c docs: finalize Session 8 â€” Fixed.md Rev 12, 59 items, 18 bugs, 4
f924841 test: wave 8 â€” btsow/torrentscsv/xfsusb/yihua (170 tests) + bt4g
6773203 test: wave 7 â€” yourbittorrent/glotorrents/pctorrent/rockbox/bittr
f191401 test: wave 6 â€” torrentfunk/torrentproject/therarbg/academicorre
1446cc5 test: wave 5 â€” audiobookbay/one337x/extratorrent (155 tests) + 3
fd543b0 docs: update CONTINUATION.md for Session 8 (4 waves, 23 plugins)
aae4069 test: wave 4 â€” kinozal/nnmclub/rutracker/megapeer/jackett + mega
e3b7acb test: wave 3 â€” rutor/tokyotoshokan/snowfl/torrentdownload/linux

```

Coverage Snapshot

Module	Coverage
api/init.py	98%
api/auth.py	91%
api/routes.py	95%
api/streaming.py	99%
main.py	94%
deduplicator.py	94%
enricher.py	100%
hooks.py	95%
jackett_autoconfig.py	99%
scheduler.py	93%
search.py	~90%
validator.py	92%
theme_injector.py	99%

Module	Coverage
env_loader.py	100%
download_proxy.py	~95%
TOTAL (unit)	88%+

Known Issues

1. **BOB-008:** RuTracker CAPTCHA “ operator-blocked (needs manual cookie paste)
 2. Go backend is a skeleton (documented in AGENTS.md)
 3. Containers may be down on session start “ bash start.sh first
 4. macOS + podman network_mode: host does NOT forward ports “ ensure-macos-tunnel.sh handles this
-

Quick-Start

```
# Pre-build gate
bash scripts/pre_build_verification.sh

# Unit tests
python3 -m pytest tests/unit/ -q --import-mode=importlib

# Coverage
python3 -m pytest tests/unit/ --cov=download-proxy/src --cov=plugins --cov=

# Lint + typecheck
ruff check . && mypy download-proxy/src/

# Frontend
cd frontend && npx vitest run

# Containers
bash start.sh && bash stop.sh
```

Architecture

Port Service	Tech
7185 qBittorrent WebUI	LinuxServer (admin/admin)
7186 Download Proxy	Python HTTP
7187 Merge Search Service	FastAPI + Angular SPA
7188 WebUI Bridge	Python (host process)
7189 boba-jackett	Go/Gin
9117 Jackett	LinuxServer (auto-configured)